

(1) A

(2) A

(3) A

(4) A

(5) D

(6) b

(7) D

2.

(a) sodium ^{hydrogen}hydrogencarbonate -1

(b) ~~phosphorus~~ ^{lead (II)} ~~di~~nitrate -1

(c) silane

(d) 1, 2 - ^{propanediol} ~~di~~propanol -1

(e) ~~carbon~~ ^{cyanide} ~~nitrite~~ ion -2

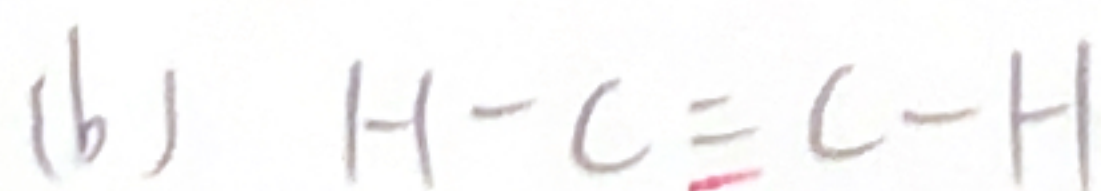
(f) sodium chlorate

(g) 2, 4 - bromomethylpentane -1

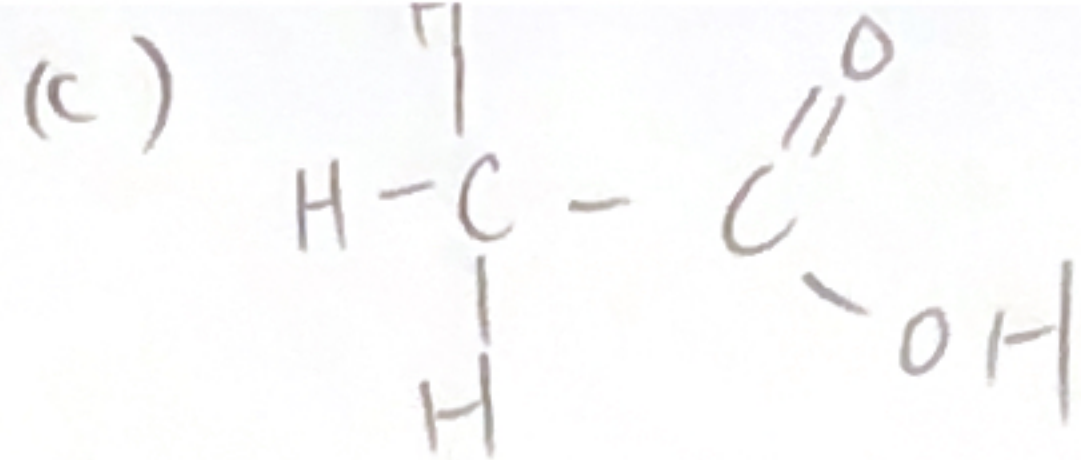
(h) 1, 3 - bromochlorobenzene -1

(i) trans - 2 - butene

3.



-1



4. $\Delta E = q + w$

$$w = -pV = -nRT = -\frac{6}{65.429} \times 8.314 \times 298 = -227.2 \text{ J} \checkmark$$

$$\Delta E = (-13980 \text{ J} + (-227.2 \text{ J})) = -14207.2 \text{ J} \checkmark$$

5. (a) $\Delta H = \frac{3}{18} \times 6.01 = 1.0016 \text{ kJ} \checkmark$

$$\Delta S = \frac{\Delta H}{T} = \frac{1.0016}{273.15} = 3.666 \text{ J/K} \checkmark$$

$$\begin{aligned} (b) \Delta S &= n C_p \ln\left(\frac{T_f}{T_i}\right) = \frac{3}{18} \times 75.3 \times \ln\left(\frac{298.15}{273.15}\right) \\ &= 1.099 \text{ J/K} + \Delta S_{\text{ice}} (3.666) \quad -2 \end{aligned}$$

$$b. (a) \Delta S = 1 \times S^\circ \text{NH}_3 - \frac{3}{2} \times S^\circ \text{H}_2 - \frac{1}{2} \times S^\circ \text{N}_2 = 1 \times 192.8 - \frac{3}{2} \times 130.7 - \frac{1}{2} \times 191.6$$

$$\begin{aligned} \Delta G^\circ &= \Delta H - T \Delta S = -45900 \times 2 - 298 \times (-99.05) \\ &= -16,36 \text{ kJ/mol} \quad \checkmark \quad -5 \end{aligned}$$

$$\begin{aligned} (b) \Delta G_{300^\circ\text{C}}^\circ &= \Delta H - T \Delta S = -45900 \times 2 - 573 \times (-99.05) \\ &= 10,85 \text{ kJ/mol} \quad \checkmark \quad -3 \end{aligned}$$

(c) $\Delta G > 0 \Rightarrow$ non spontaneous $\checkmark$

(d) No, with the temperature increase, ΔG also increase.
 $\Rightarrow$ non spontaneous $\checkmark$

1.

$$(a) \Delta G^\circ = -RT \ln K_p = -371^{000}_{(kJ)} - 8.314 \times 298 \times \ln K_p$$

$$\Rightarrow K_p = 1.164 \quad \# \quad \times \quad -2$$

$$(b) \Delta G^\circ = \Delta G_f^\circ \text{PSCl}_3 - \Delta G_f^\circ \text{PCl}_3$$

$$= -272.4 - (-363.2) = 90.8 \text{ kJ/mol} \times$$

90800 J/mol.

$$= -RT \ln K_p = -8.314 \times 298 \times \ln K_p \quad -3$$

$$\Rightarrow K_p = 0.964 \quad \# \quad \times$$

$$8. (a) \Delta G^\circ = -\Delta G_f^\circ (\text{Cu}_2\text{O}) = 146 \text{ kJ/mol} \quad \checkmark$$

146000 J/mol.

$$\Delta G^\circ = -RT \ln K_p = -8.314 \times 298 \times \ln K_p$$

$$K_p = 0.942 \quad \# \quad \times \quad -2$$

$$(b) \Delta G^\circ = \Delta G_f^\circ (\text{CO}) - \Delta G_f^\circ (\text{Cu}_2\text{O}) = -137.2 - (-146) = 8.8 \text{ kJ/mol} \quad \checkmark$$

8800 J/mol

$$\Delta G^\circ = -RT \ln K_p = -8.314 \times 298 \times \ln K_p$$

$$K_p = 0.996 \quad \# \quad \times \quad -2$$